$$L_{\mathrm{AE}} = L_{\mathrm{rec}} + \lambda_{\mathrm{adv}} L_{\mathrm{adv}} + L_{\mathrm{reg}}$$

$$L_{\mathrm{rec}} : \ \text{pixel + perceptual reconstruction}$$

$$L_{\mathrm{reg}} : \ \text{weak KL or VQ regularization}$$